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Carcinoma of pancreas

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2025

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CARCINOMA OF THE PANCREAS

- Pancreatic cancer is the sixth leading cause of cancer death in the UK, and the incidence is 10 cases per 100 000 population per year.
- Worldwide, it constitutes 2–3% of all cancers and, in the USA, is the fourth highest cause of cancer death.
- There is no simple screening test; however, patients with an increased inherited risk of pancreatic cancer (Table 1) should be referred to specialist units for screening and counseling

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TABLE 1 **Risk factors for the development of pancreatic cancer.**
Demographic factors
Age (peak incidence 65–75 years)
Male gender
Black ethnicity
Environment/lifestyle
Cigarette smoking
Genetic factors and medical conditions
Family history
Two first-degree relatives with pancreas cancer: relative risk increases 18- to 57-fold
Germline BRCA2 mutations in some rare high-risk families
Hereditary pancreatitis (50- to 70-fold increased risk)
Chronic pancreatitis (5- to 15-fold increased risk)
Lynch syndrome (HNPCC)
Ataxia telangiectasia
Peutz–Jeghers syndrome
Familial breast–ovarian cancer syndrome
Familial atypical multiple mole melanoma
Familial adenomatous polyposis – risk of ampullary/duodenal carcinoma
Diabetes mellitus

Pathology

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- More than 85% of pancreatic cancers are ductal adenocarcinomas.
- The remaining tumors constitute a variety of pathologies with individual characteristics. Usually solid .
- Endocrine tumors of the pancreas are rare
- Ductal adenocarcinomas arise most commonly in the head of the gland.
- Ductal adenocarcinomas infiltrate locally, typically along nerve sheaths, along lymphatics and into blood vessels.
- Liver and peritoneal metastases are common

- Cystic tumours of the pancreas may be serous or mucinous.
- **Serous cystadenomas** are typically found in older women,They are benign.
- **Mucinous tumours**, on the other hand, have the potential for malignant transformation.
- They include mucinous cystic neoplasms (MCNs) and intraductal papillary mucinous neoplasms (IPMNs).
- Tumours arising from the ampulla or from the distal common bile duct can present as a mass in the head of the pancreas, and constitute around a third of all tumours in that area(periampullary cancer)

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Clinical features

- **Jaundice** secondary to obstruction of the distal bile duct is the most common symptom that draws attention to ampullary and pancreatic head tumors.
- it is characteristically **painless jaundice** but may be associated with nausea and epigastric discomfort.
- **Pruritus, dark urine and pale stools** with steatorrhoea are common accompaniments of jaundice.
- In the absence of jaundice, symptoms are often non-specific, namely vague discomfort, anorexia and weight loss, and are frequently dismissed by both patient and doctor.

- Upper abdominal symptoms in a **recently diagnosed diabetic**, especially in one above 50 years of age, with no family history or obesity, should raise suspicion.
- unexplained attack of pancreatitis; all such patients should have follow-up imaging of the pancreas.
- Tumours of the body and tail of the gland often grow silently, and present at an advanced unresectable stage.

- **Back pain** is a worrying symptom, raising the possibility of retroperitoneal infiltration.
- On examination, there may be evidence of jaundice, weight loss, a palpable liver and a palpable gall bladder. **Courvoisier** first drew attention to the association of an enlarged gall bladder and a pancreatic tumour in 1890, when he noted that, when the common duct is obstructed by a stone, distension of the gall bladder (which is likely to be chronically inflamed) is rare; when the duct is obstructed in some other way, such as a neoplasm, distension of the normal gall bladder is common.
- palpable mass, ascites, supraclavicular nodes and tumour deposits in the pelvis; when present, they indicate a grim prognosis.

Investigation

In a jaundiced patient,:

- usual blood tests
- ultrasound scan determine if the bile duct is dilated. If it is, and there is a genuine suspicion of a tumour in the head of the pancreas,
- the preferred test is a contrast-enhanced CT scan.
- The presence of hepatic or peritoneal metastases, lymph node metastases distant from the pancreatic head, or encasement of the superior mesenteric, hepatic or coeliac artery by tumour are clear **contraindications to surgical resection**.
- Tumour size, continuous invasion of the duodenum, stomach or colon, and lymph node metastases within the operative field are **not contraindications.**

MRI and MR angiography can provide information comparable to CT. **double duct sign**

ERCP and biliary stenting should be carried out :

- --if there is any suggestion of cholangitis,
- -- if there is diagnostic doubt (small ampullary lesions may not be seen on CT, and ERCP is the best way to identify them) or
- -- if there is likely to be a delay between diagnosis and surgery and the patient is deeply jaundiced with distressing pruritus. It relieves the jaundice and can also
- -- provide a brush cytology or biopsy specimen to confirm the diagnosis. Otherwise, however, preoperative ERCP and biliary stenting is not mandatory in patients with resectable disease.

The prothrombin time should be checked, and clotting abnormalities should be corrected with vitamin K or fresh-frozen plasma prior to ERCP.

- **EUS** is useful
- if CT fails to demonstrate a tumour,
- if tissue diagnosis is required prior to surgery (e.g. a mass has developed on a background of chronic pancreatitis and a distinction needs to be made between inflammation and neoplasia),
- if vascular invasion needs to be confirmed,
- or in separating cystic tumours from pseudocysts.

Transduodenal or transgastric FNA or Trucut biopsy performed under EUS guidance avoids spillage of tumourcells into the peritoneal cavity.

- **Percutaneous transperitoneal biopsy** of potentially resectable pancreatic tumours should be avoided as far as possible.
- **Histological confirmation of malignancy** is desirable but not essential, particularly if the imaging clearly demonstrates a resectable tumour.
- **Diagnostic laparoscopy** prior to an attempt at resection can spare a proportion of patients an unnecessary laparotomy by identifying small peritoneal and liver metastases.
- It can be combined with **laparoscopic ultrasonography**.
- The **tumour marker CA19-9** is not highly specific or sensitive, but a baseline level should be established; if it is initially raised, it can be useful later in identifying recurrence.

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Management

- At the time of presentation, more than 85% of patients with ductal adenocarcinoma are unsuitable for resection because the disease is too advanced.
- If imaging shows that the tumour is potentially resectable, the patient should be considered for surgical resection, as that offers the only (albeit small) chance of a cure.

- Comorbidities should be taken carefully into account.
- Biological rather than chronological age should be the consideration.
- • **If a cystic tumour** is encountered, no matter how large, surgical resection should be considered, as it carries a reasonable chance of cure.
- • **Tumours of the ampulla** have a good prognosis and should, if at all possible, be resected.
- • Some of the rare tumours and the **neuroendocrine** lesions should also be resected if at all possible.
- For those patients who have inoperable disease, palliative treatment should be offered.

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Surgical resection

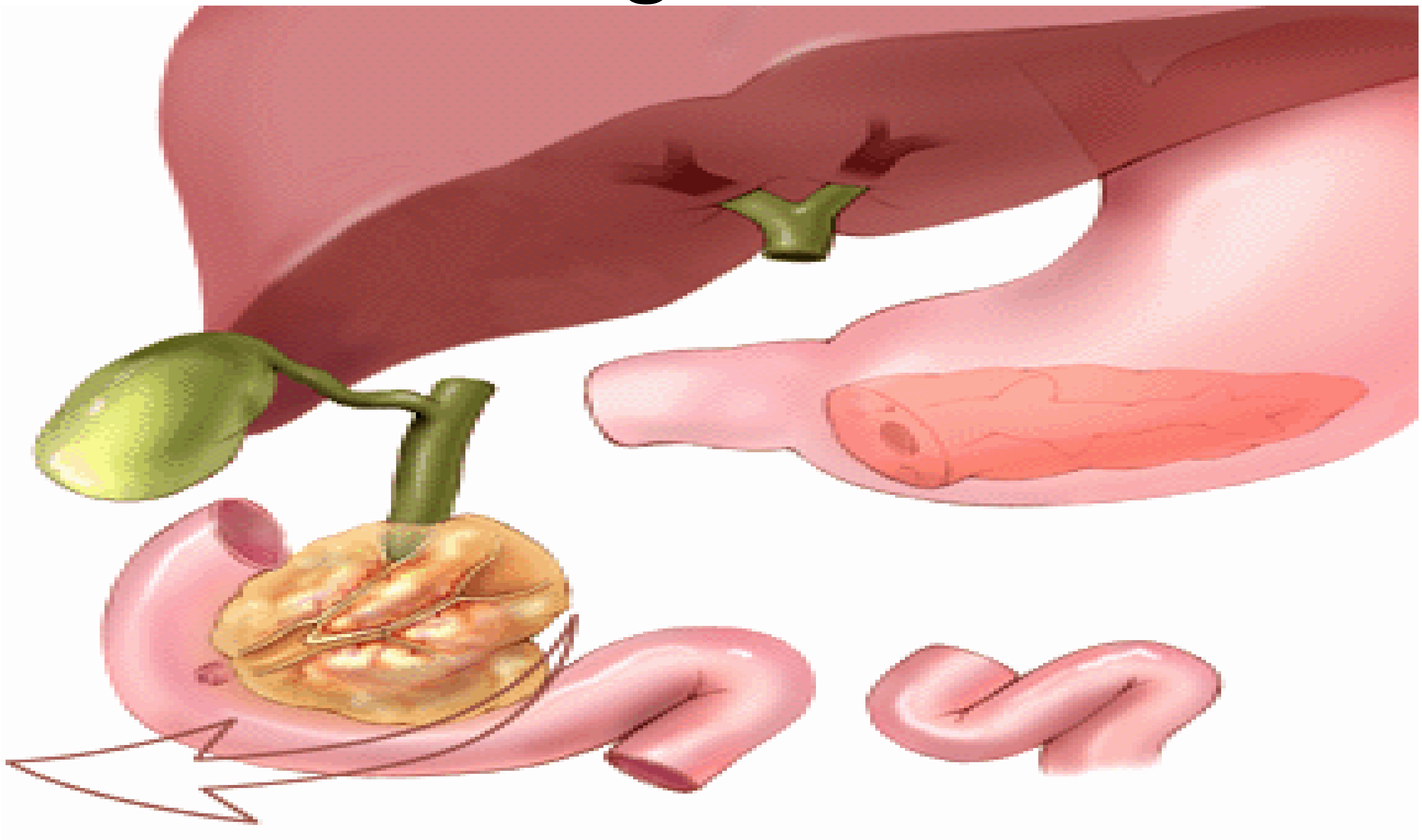
- The standard resection for a tumour of the pancreatic head or the ampulla is a **pylorus-preserving pancreatoduodenectomy (PPPD**). This involves removal of the duodenum and the pancreatic head, including the distal part of the bile duct. (figure 1 and 2)
- The **original pancreatoduodenectomy** as proposed **by Whipple** included resection of the gastric antrum.

Preserving the antrum and the pylorus is thought to result in a more physiological outcome with no difference in survival or recurrence rates.

The Whipple procedure is now reserved for situations in which the entire duodenum has to be removed (e.g. in FAP) or where the tumour encroaches on the first part of the duodenum or the distal stomach and a PPPD would not achieve a clear resection margin.

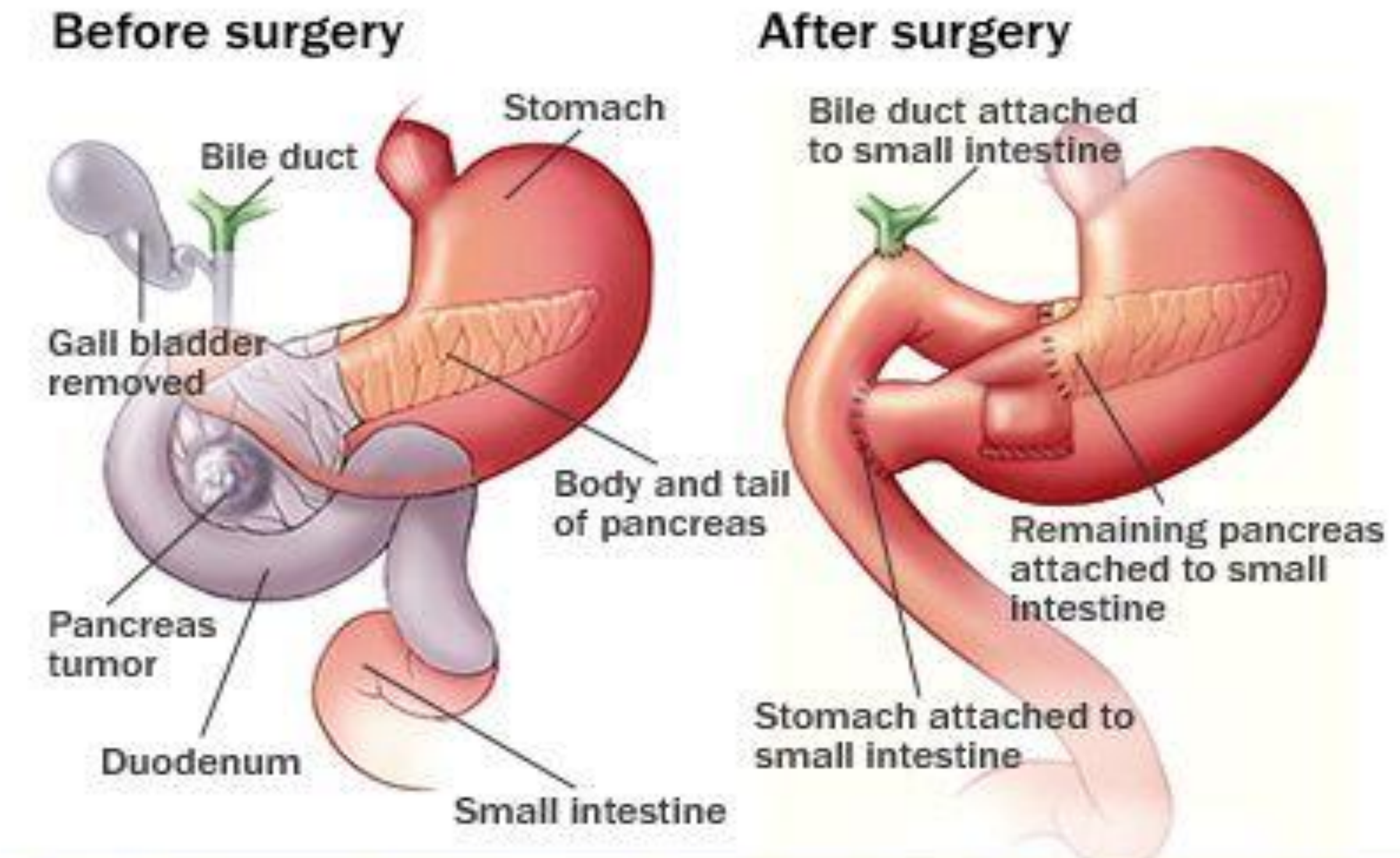
The PPPD procedure includes a local lymphadenectomy

Figure 1



Organs removed during a Whipple

Figure 2



Before surgery
After surgery
Stomach
Bile duct
Gall bladder removed
Body and tail of pancreas
Pancreas tumor
Duodenum
Small intestine
Bile duct attached to small intestine
Remaining pancreas attached to small intestine
Stomach attached to small intestine

- Video

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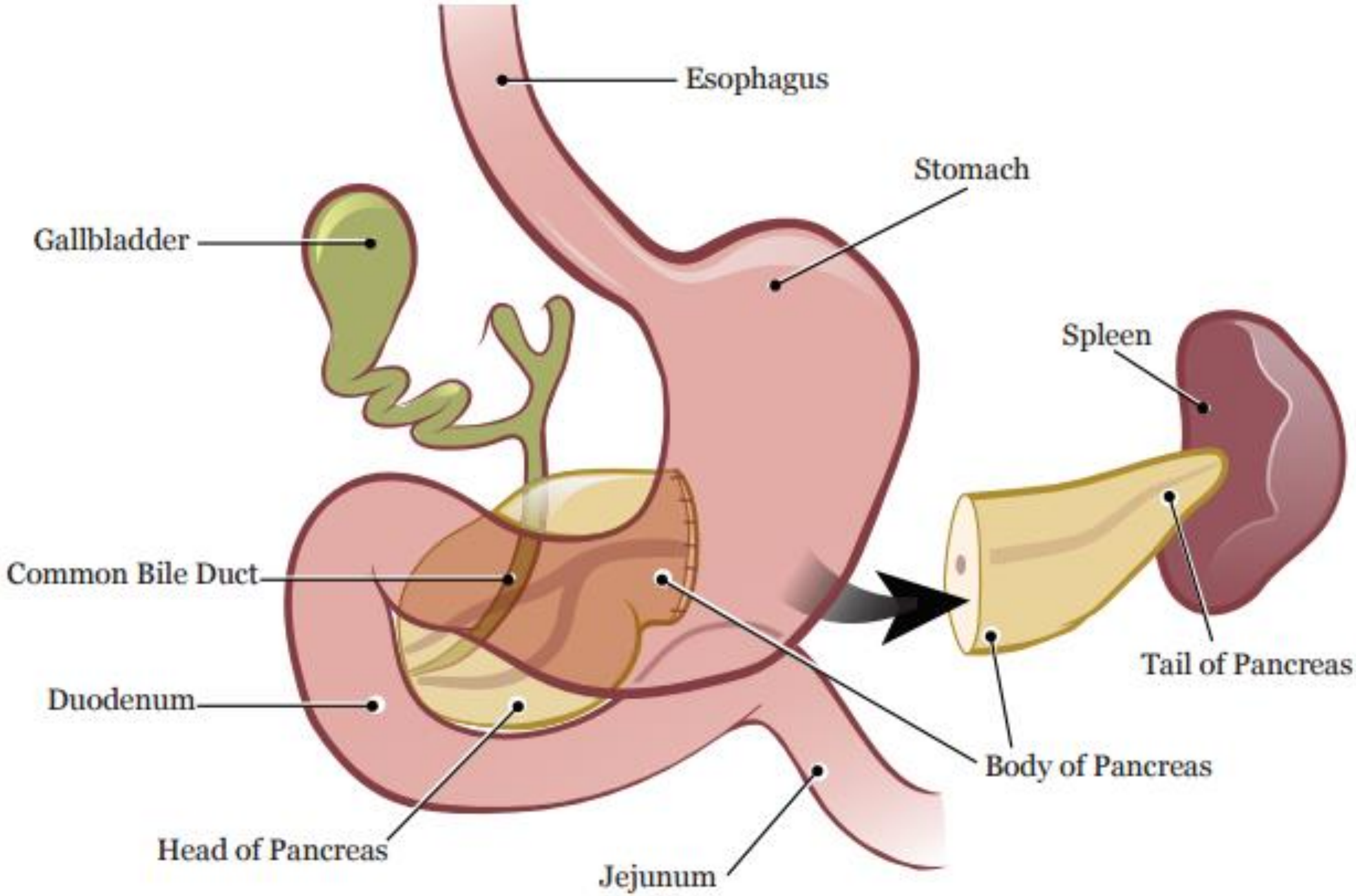
- **Total pancreatectomy** is warranted only in situations where one is dealing with a multifocal tumour (e.g. a main duct IPMN), or the body and tail of the gland are too inflamed or too friable to achieve a safe anastomosis with the bowel..

- For tumours of the body and tail, **distal pancreatectomy with splenectomy** is the standard.
- Infiltration of the splenic artery or vein by the tumour is not a contraindication to resection.(figure 3)
- When resecting the pancreatic tail for a benign lesion, one may attempt to preserve the spleen if possible. When removing the spleen, **prior vaccinations** against pneumococci, meningococci and Haemophilus influenzae B should be administered, and subsequent antibiotic prophylaxis given .

Figure 3


Esophagus
Stomach
Gallbladder
Spleen
Common Bile Duct
Tail of Pancreas
Duodenum
Body of Pancreas
Head of Pancreas
Jejunum

- While the majority of pancreatic resections continue to be performed via the open approach, there is evidence that the laparoscopic and robotic approaches are also feasible and may yield comparable results.

PANCREATODUODENECTOMY

- • The patient’s coagulation screen should be checked preoperatively and
- • adequate hydration ensured.
- • The patient should be aware of the diagnosis, the gravity of the operation and the risks involved.

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The operation has three distinct phases:

- exploration and assessment;
- resection;
- reconstruction.
- The operation should take between 3 and 6 hours.
- Blood loss should be low, and transfusion is often not necessary.
- The patients are usually nursed in a high-dependency area for the first 24–48 hours after surgery.
- Prolonged nasogastric drainage is unnecessary, and early feeding can be commenced.
- Resection for pancreatic cancer should be carried out in specialist units.

Complications of surgery

- PPPD should carry a mortality of no more than 3–5%.
- The morbidity remains high, with some 30–40% of patients developing a complication in the postoperative period.
- These complications are usually **infective**, but a **leak** from the anastomosis between the pancreas and the bowel is known to occur in at least 10% of patients, and this may give rise to major complications. Octreotide may be administered in the perioperative period to suppress secretion and reduce the likelihood of a leak.

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ADJUVANT THERAPY

- Most patients with resected ductal adenocarcinoma are now offered 6 months of adjuvant chemotherapy with gemcitabine and/or 5-FU.
- Some centres continue to offer chemoradiotherapy, particularly in patients with involved (R1) resection margins.
- Patients with resected ampullary tumours have a 5-year survival of 40%, and cystic tumours and neuroendocrine tumours can often be cured by surgical resection.

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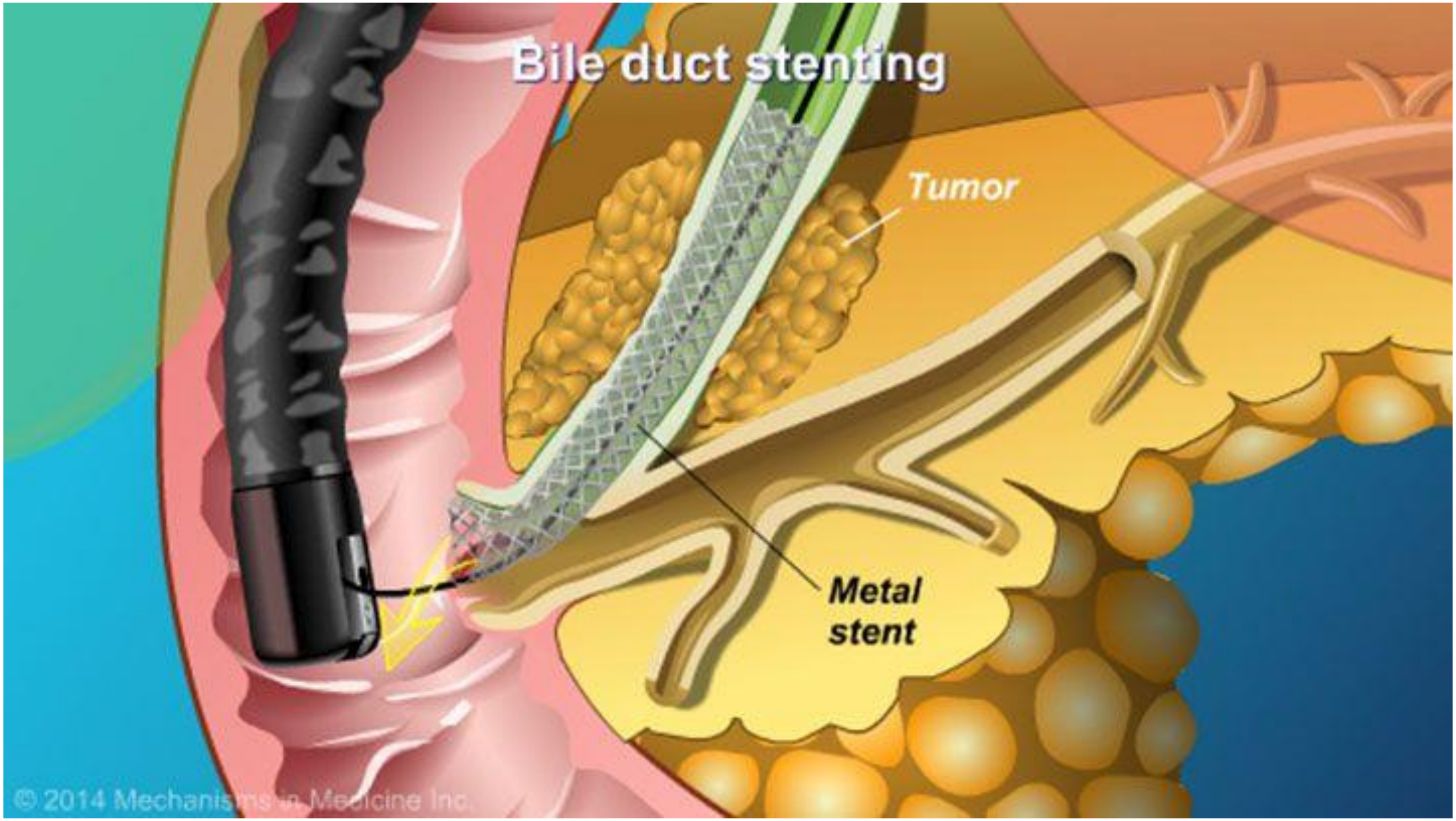
Palliation

- The median survival of patients with unresectable, locally advanced, non-metastatic pancreatic cancer is 6–10 months and, in patients with metastatic disease, it is 2–6 months.
- If unresectable disease is found in the course of a laparotomy that was commenced with the intent to resect, a chole dochoenterostomy and a gastroenterostomy should be carried out to relieve (or pre-empt) jaundice and duodenal obstruction.
- The bile duct may be anastomosed to the duodenum, or to a loop of jejunum. It is preferable to use the bile duct rather than the gall bladder. Cholecystojejunostomy is easier to perform, but the bile must then drain through the cystic duct, which is narrow and, if inserted low into the bile duct, is vulnerable to occlusion by tumour growth.
- A coeliac plexus block can also be administered.
- A transduodenal Trucut biopsy of the tumour should be obtained.

- In patients found to have unresectable disease on imaging, jaundice is relieved by stenting at ERCP (Figure 4).
- Stents may be made of plastic or self-expanding metal mesh.
- Plastic stents are cheaper but tend to occlude faster and, if the patient is likely to have a longer life expectancy, a metal stent can be used.

Figure 1


Bile duct stenting
Tumor
Metal
stent
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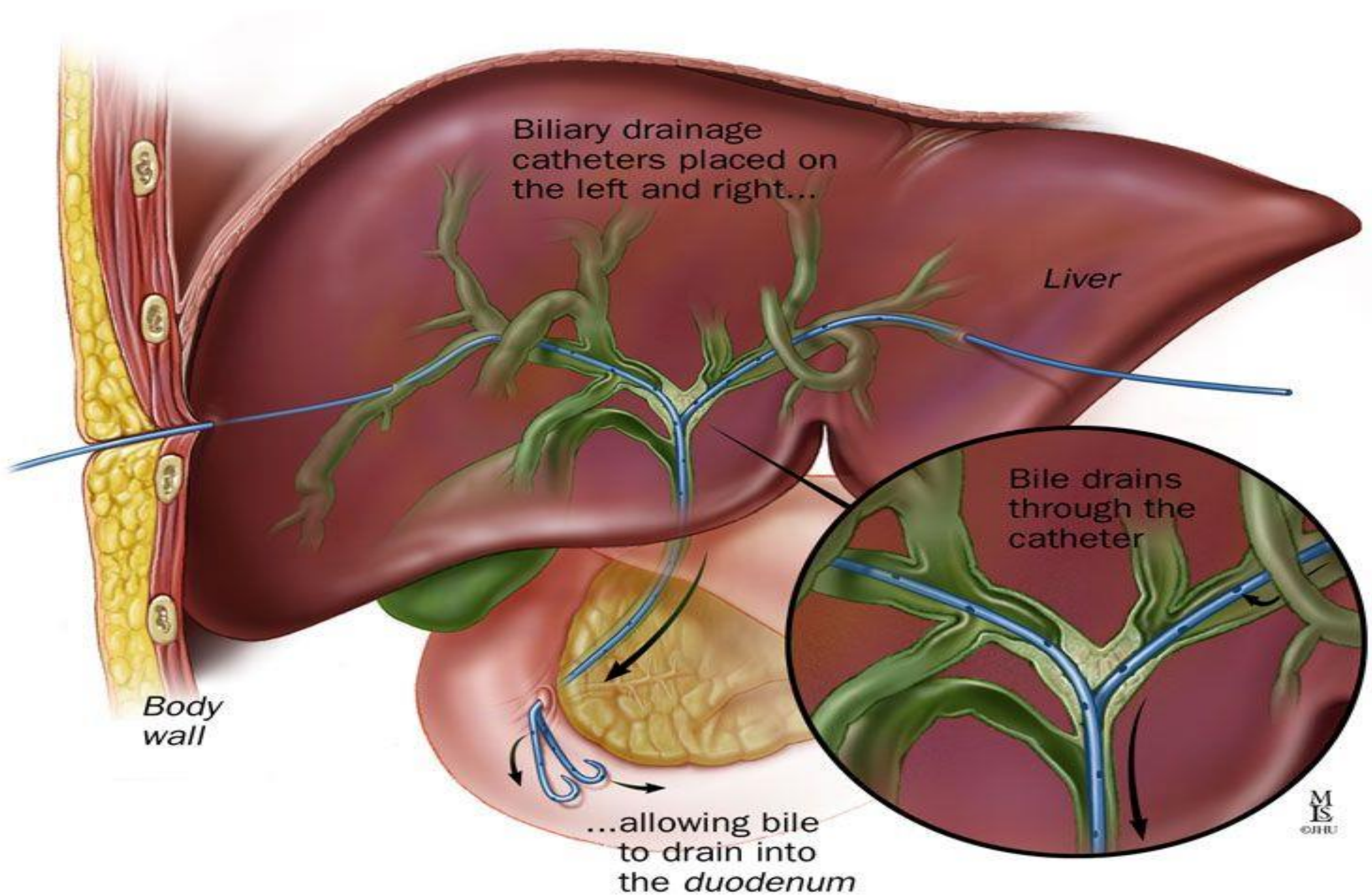
If the patient is not a suitable candidate for endoscopic biliary stenting, a percutaneous transhepatic stent can



- Obstruction of the duodenum occurs in approximately 15% of cases.
- If this occurs **early** in the course of the disease, surgical bypass by gastrojejunostomy is appropriate but,
- if it is **late** in the course of the disease, then the use of expanding metal stents inserted endoscopically is preferable, as many of these patients have prolonged delayed gastric emptying following surgery (Figure 6).
- If both biliary and duodenal metal stents are to be placed endoscopically, the biliary one should be placed first.

Figure 6


A
B
Enteral stent
in duodenum
x-ray image
Self-expanding metal stent
restores patency in duodenum

- If no operative procedure is undertaken, an EUS-guided or percutaneous biopsy of the tumour should be performed before consideration of chemotherapy or chemoradiation.
- Lymphomas of the pancreas are rare and constitute less than 3% of all pancreatic cancers. These respond to chemoradiotherapy and surgical resection is not indicated.
- For patients with ductal adenocarcinoma, 5-FU or gemcitabine will produce a remission in 15–25%, while the remainder will receive no benefit from the therapy.

- No long-term cures have been described with chemotherapy or radiotherapy.
- Attempts to downstage unresectable disease with chemotherapy or chemoradiation and render it resectable are rarely successful
- Steatorrhoea is treated with enzyme supplementation.
- Diabetes mellitus, if it develops, is treated with oral hypoglycaemics or insulin as appropriate, and pain with either analgesics or an appropriate nerve block.

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Palliation of pancreatic cancer

- **Relieve jaundice and treat biliary sepsis**
  - ●● Surgical biliary bypass
  - ●● Stent placed at ERCP or percutaneous transhepatic cholangiography
- **Improve gastric emptying**
  - ●● Surgical gastroenterostomy
  - ●● Duodenal stent
- **Pain relief**
  - ●● Stepwise escalation of analgesia
  - ●● Coeliac plexus block
  - ●● Transthoracic splanchnicectomy
- **Symptom relief and quality of life**
  - ●● Encourage normal activities
  - ●● Enzyme replacement for steatorrhoea
  - ●● Treat diabetes
- **Consider chemotherapy**

Bailey & Love’s SHORT PRACTICE of SURGERY
28th EDITION 2023.